Co-NAML-LSTUR: A Combined Model with Attentive Multi-View Learning and Long- and Short-term User Representations for News Recommendation

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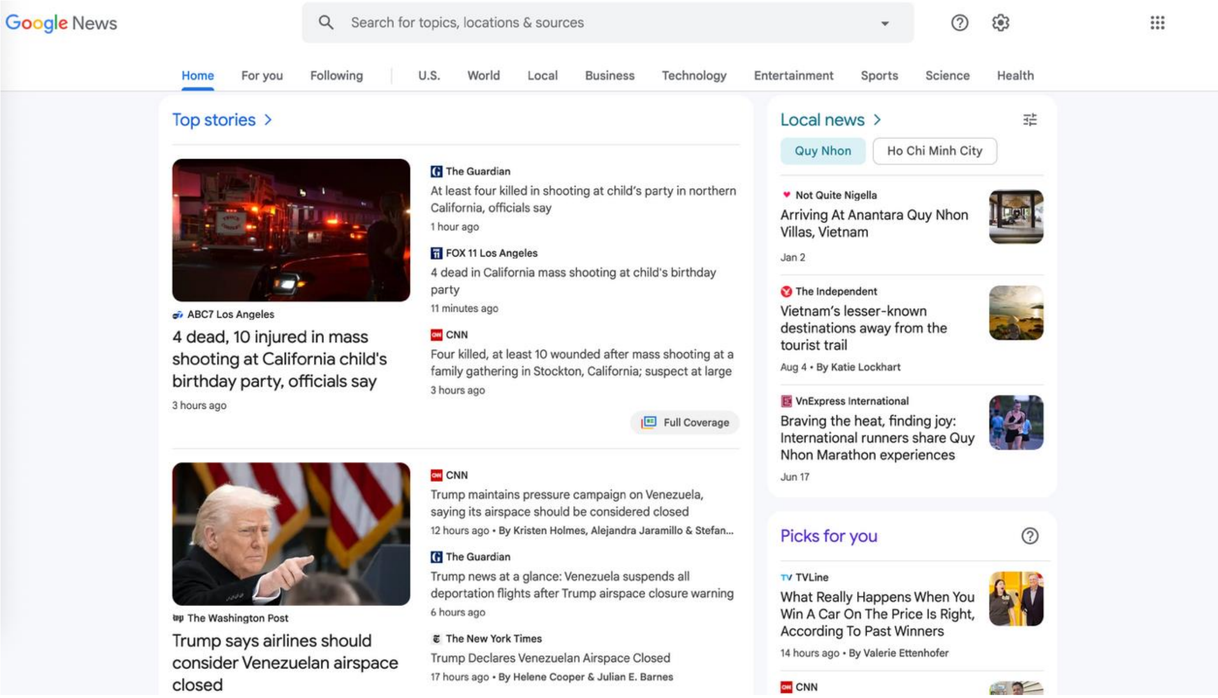
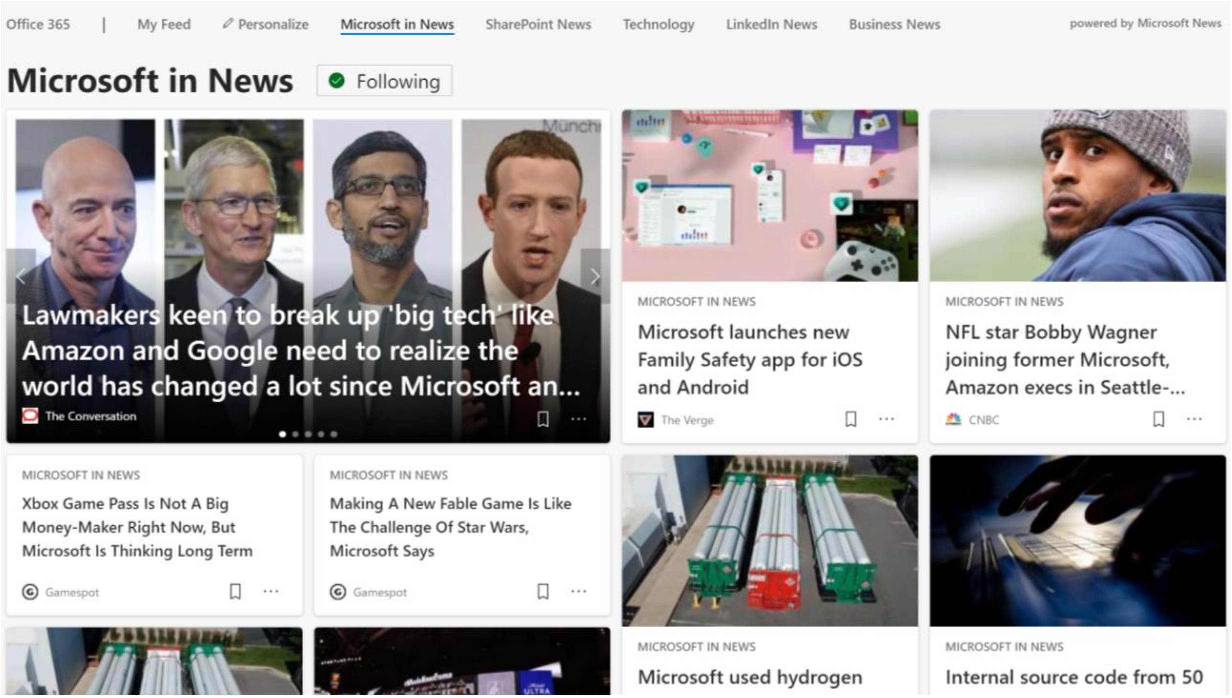
Dec 05, 2025

What is News Recommendation?

- **Online News platforms** become **popular** for people to read News.
- **Personalized News recommendation** is **important** for online News platforms.



News in Google News [1]



News in Microsoft News [2]

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[1] https://news.google.com
[2] https://microsoftnews.msn.com

Personalized News Recommendation

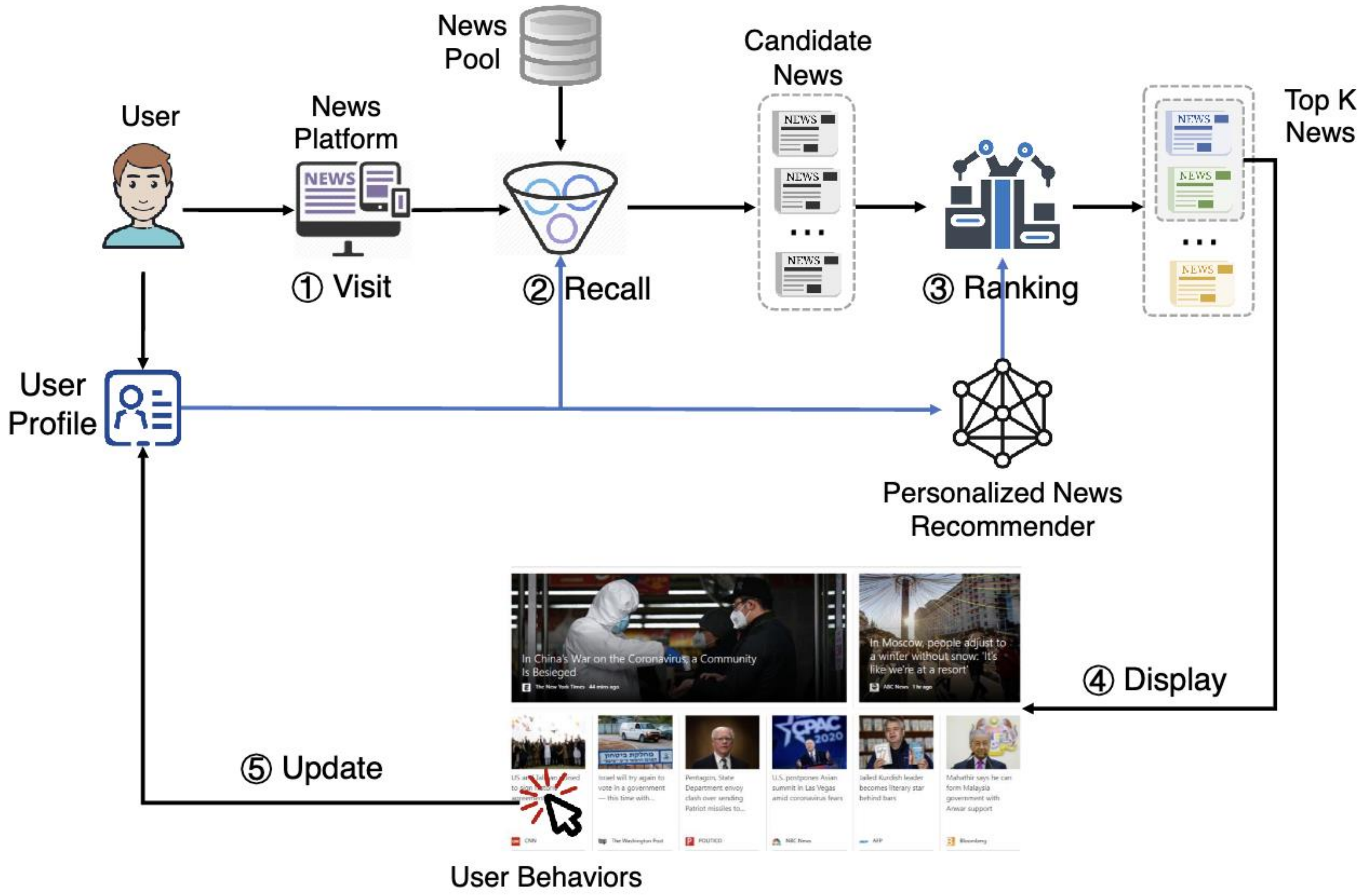
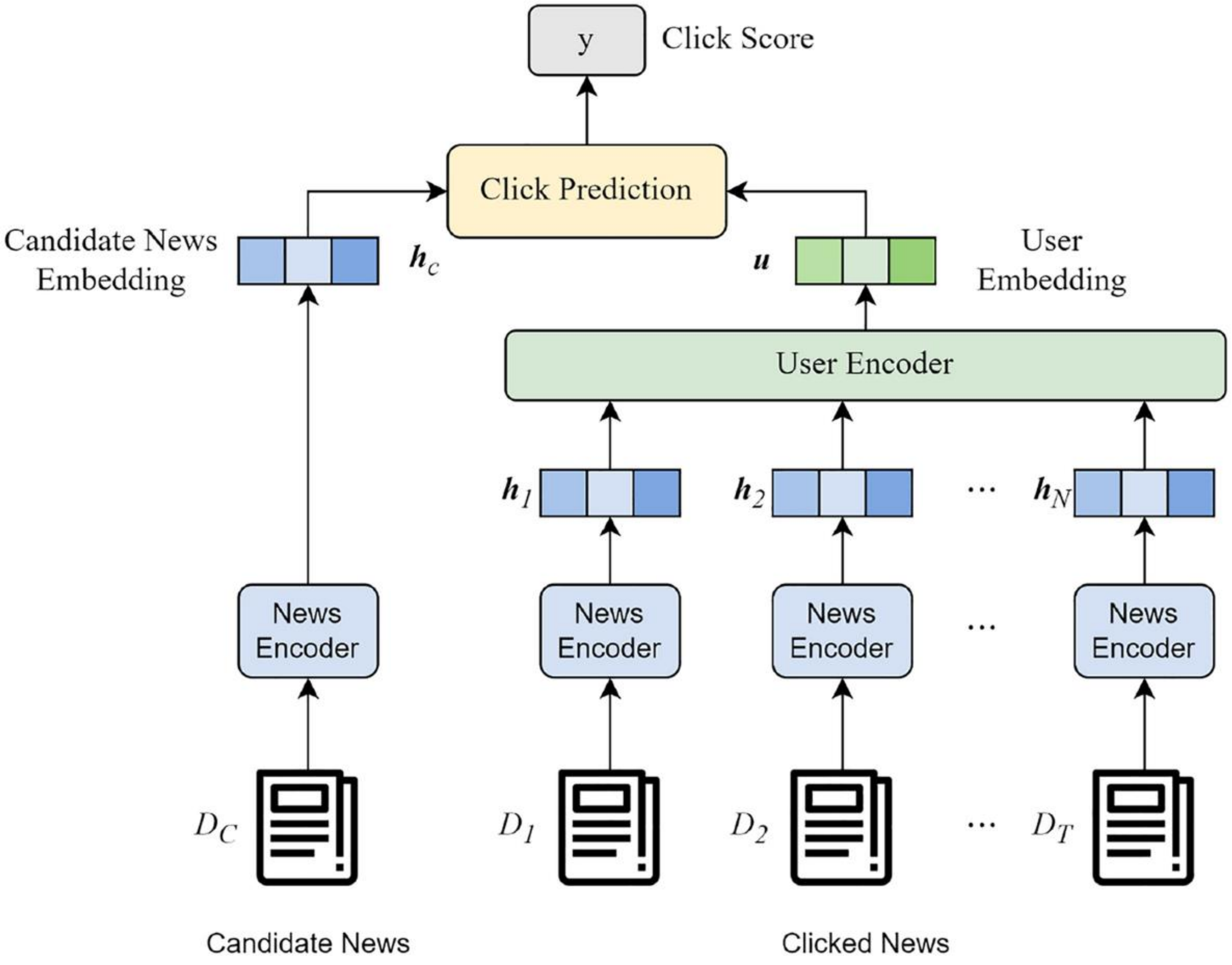


Fig. 1. An example workflow of personalized news recommender systems. [3]

Personalized News recommendation aims to **deliver relevant articles** based on **users' personal interest**.

[3] Wu, Chuhan, et al. "Personalized news recommendation: Methods and challenges." ACM Transactions on Information Systems 41.1 (2023): 1-50.

Personalized News Recommendation



The Framework of mainstream personalized news recommendation system. [4]

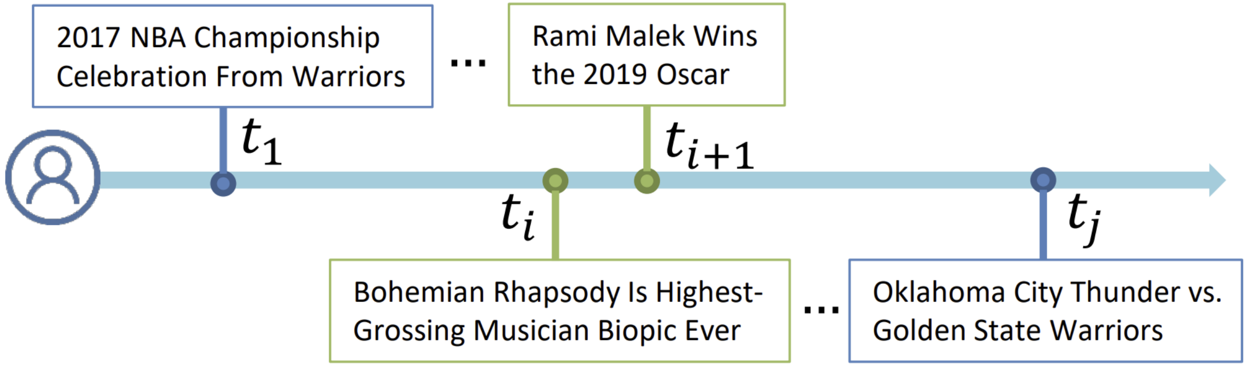
- **User interest modeling** is the **core task** of news recommendation.
- Steps:
  - **News encoder** and **User encoder** extract the features of **news articles** and **user interests.**
  - **The model ranks** the **candidate news** according to the **matching** between the information of **News** and **User's interest**.
  - The **higher** the **score/probability** of **user clicking** on the candidate news

[4] Meng, Xiangfu, et al. "A survey of personalized news recommendation." Data Science and Engineering 8.4 (2023): 396-416.

Challenges and Goal

Category	Sports	Entertainment
Title	Astros improve outfield, agree to 2-year deal with Brantley	The best games of 2018
Body	Outfielder Michael Brantley agreed to a two-year, $32 million contract with Houston, sources familiar with the deal told Yahoo Sports, bringing his steady left-handed bat to the top of an Astros ...	The Best Games of 2018 Superheroes, super-dads, and Super Mario parties brought the joy in 2018 to millions of players who ate up the year's impressive achievements in gaming ...

Examples of TWO user clicked News with different usage words. [5]



An example of long-term and short-term user interests in news reading. [6]

Challenges

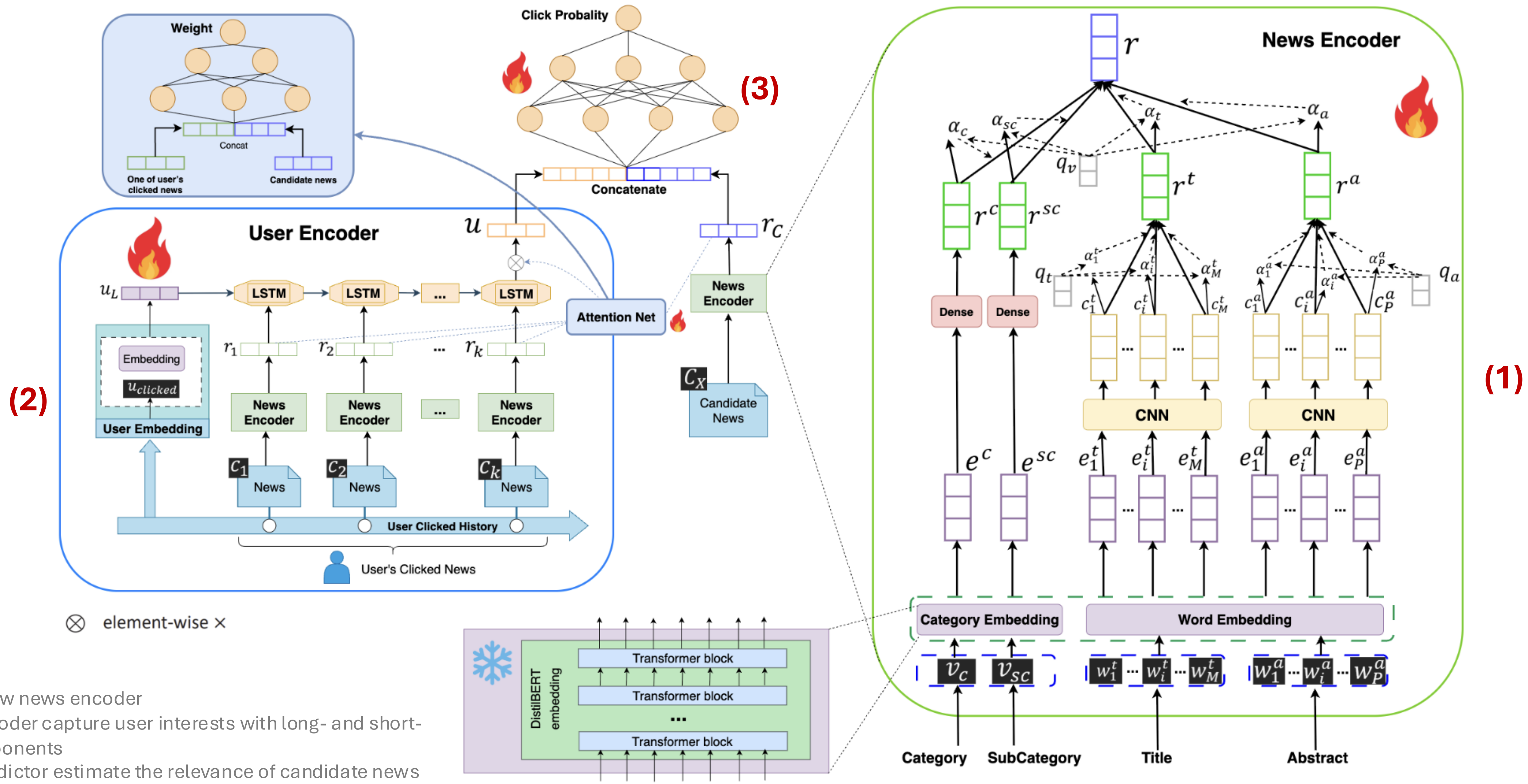
➡ Uneven word importance inside news text.
➡ Dual-scale user interest (long-term vs short-term).
➡ Lack of research on jointly handling both cases.

Goal

Build a unified news-recommendation representation that (1) **understands word-level importance** and (2) models **both long-term and short-term user interests**.

[5] Wu, Chuhan, et al. "Neural news recommendation with attentive multi-view learning." arXiv preprint arXiv:1907.05576 (2019).
[6] An, Mingxiao, et al. "Neural news recommendation with long-and short-term user representations." Proceedings of the 57th annual meeting of the association for computational linguistics. 2019.

Co-NAML-LSTUR overview

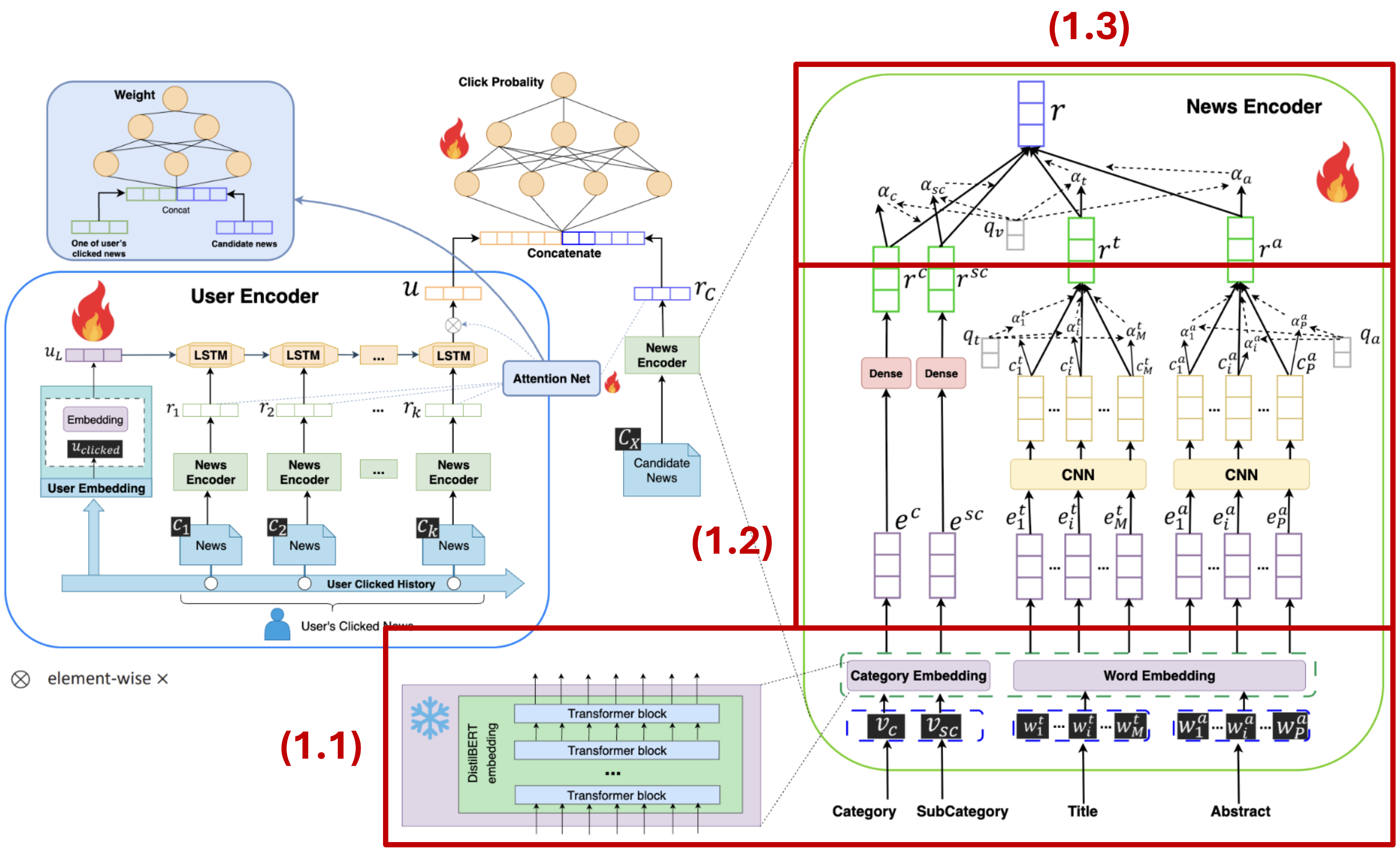


(1) A multi-view news encoder
(2) A user encoder capture user interests with long- and short-term components
(3) A click predictor estimate the relevance of candidate news

The implementation of our model is publicly available at: https://github.com/MinhNguyenDS/Co-NAML-LSTUR

Co-NAML-LSTUR

News Encoder



Attentive Multi-view Learning to capturing complementary signals from different textual and categorical components of news.

Linguistic representations. We adopt **DistilBERT** [7] as the underlying encoder.

- Four types of information: **Category**, **SubCategory, Title**, and **Abstract**.

Capture local semantic patterns. We use:

- **Nonlinear projections** or
- **1D CNN** [8] layer & **Word-level attention** [9]

$$r^c = \mathrm{ReLU}(V_c \cdot \mathrm{Emb}(v_c) + b_c)$$

$$c_i^t = \mathrm{ReLU}(F_t \cdot e^t_{(i-k):(i+k)} + b_t)$$

$$a_i^t = q_t^\top \tanh(V_t c_i^t + v_t), \quad \alpha_i^t = \frac{\exp(a_i^t)}{\sum_{j=1}^{M} \exp(a_j^t)} \quad r^t = \sum_{j=1}^{M} \alpha_j^t c_j^t$$

View-level Attention. We use a **View-level attention**:

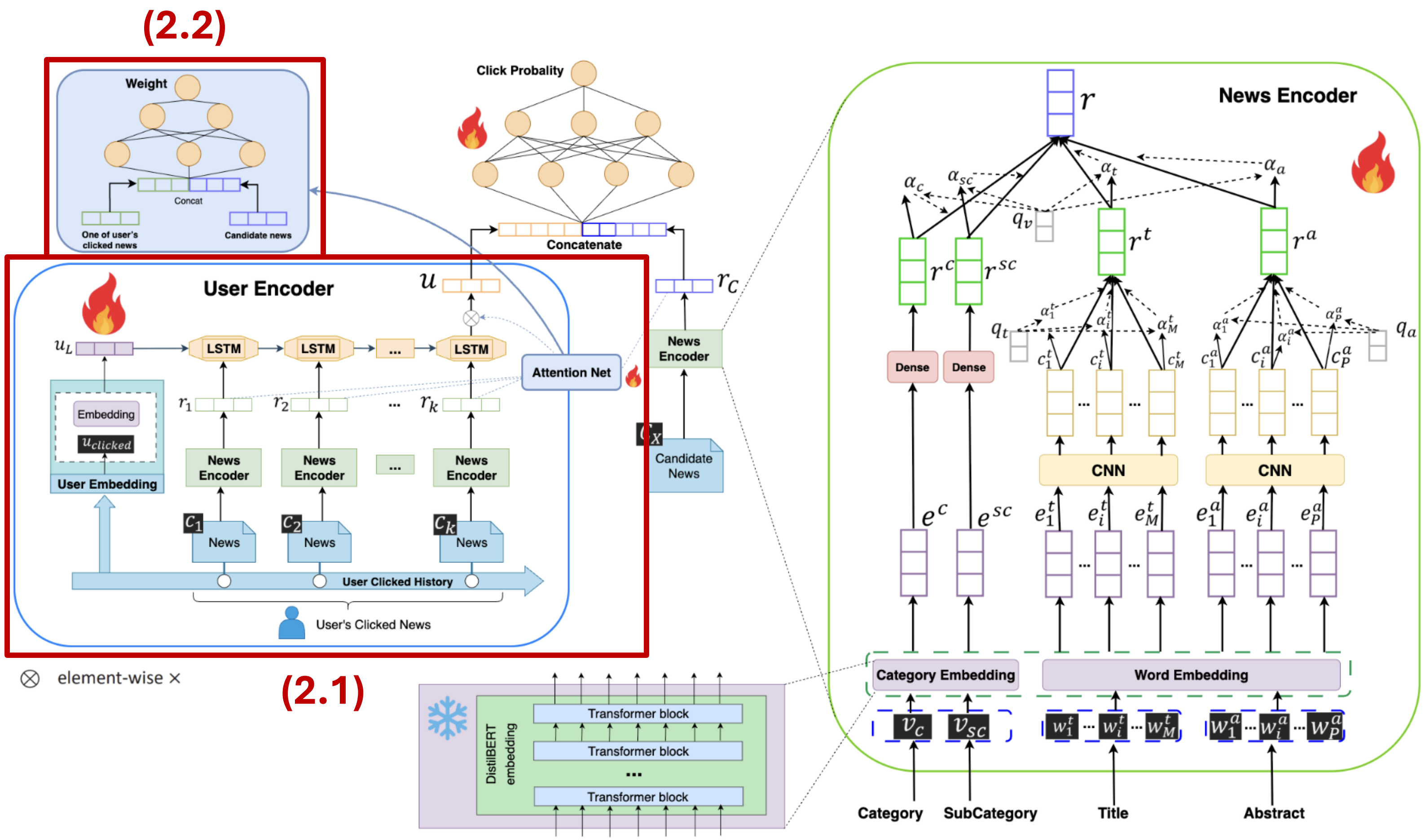
$$a_t = q_v^\top \tanh(U_v r^t + u_v), \quad \alpha_t = \frac{\exp(a_t)}{\sum_{v \in \{t,abs,c,sc\}} \exp(a_v)}$$

$$r = \alpha_t r^t + \alpha_{abs} r^{abs} + \alpha_c r^c + \alpha_{sc} r^{sc}$$

[7] Sanh, Victor, et al. "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter." arXiv preprint arXiv:1910.01108 (2019).
[8] Kim, Yoon. "Convolutional neural networks for sentence classification." arXiv preprint arXiv:1408.5882 (2014).
[9] Wu, Chuhan, et al. "Neural demographic prediction using search query." Proceedings of the twelfth ACM international conference on web search and data mining. 2019.

Co-NAML-LSTUR

User Encoder + Attention-based User Interest Extraction



User's recent reading history as a time-ordered sequence $C = \{c_1, c_2, \ldots, ck\}$

Long-Term User Representation. Follow LSTUR's [10] strategy of learning user embeddings via user IDs.

$$u_l = W_u[u]$$

Short-Term User Representation. We use LSTM [11] for capturing longer user histories dependencies.

$$u_s = \mathrm{LSTM}(r_1, r_2, \ldots, r_k)$$

Attention-based User Interest Extraction. Enhance the expressiveness of user embeddings *(user preferences are often diverse and context-dependent)*

$$s_i = \frac{\exp(H([r_i; r_c]))}{\sum_{j=1}^{k} \exp(H([r_j; r_c]))} \qquad u_{att} = \sum_{i=1}^{k} s_i \cdot r_i$$

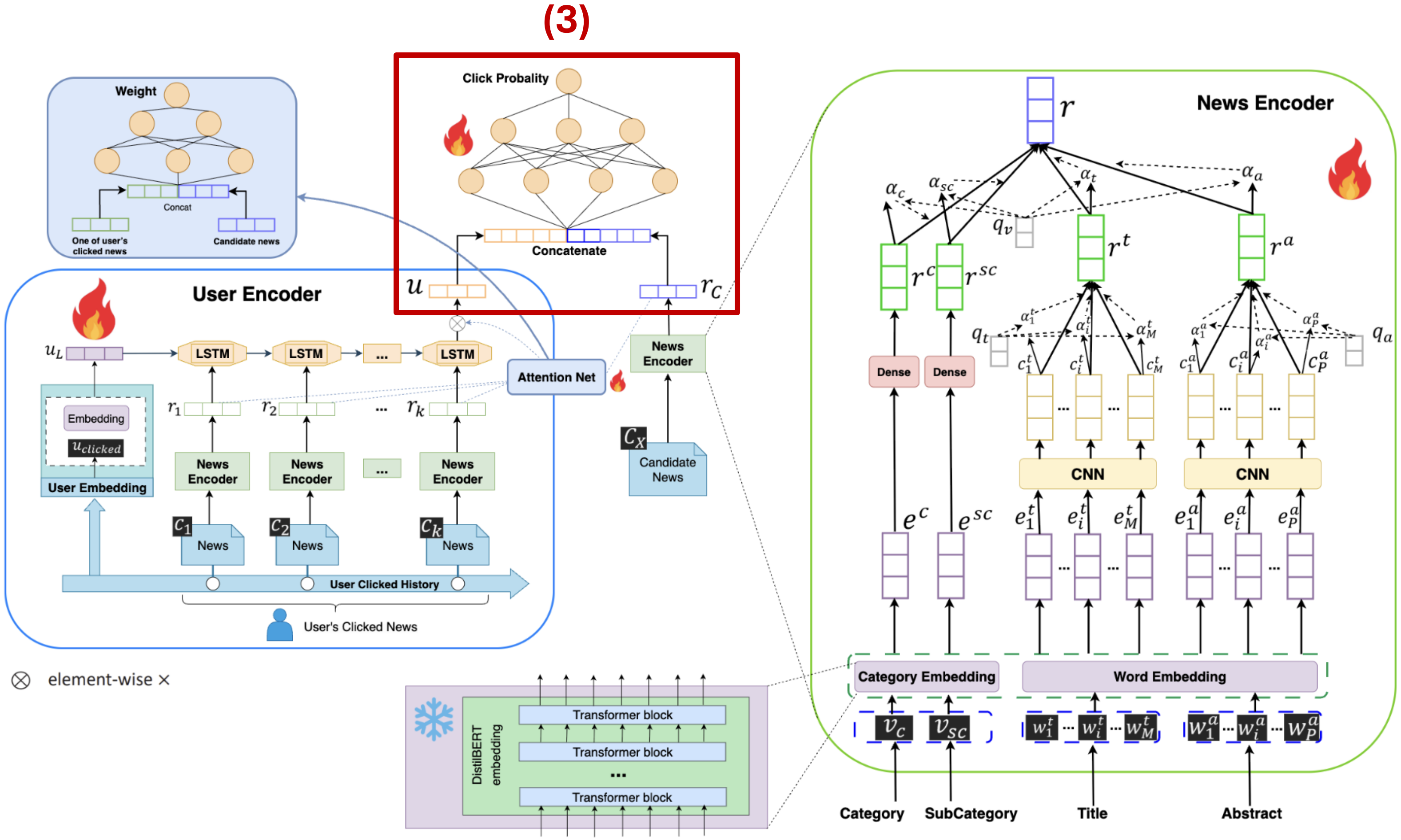
Integration of Long- and Short-Term Representations. We adopt the **LSTUR-ini** strategy

$$u = \mathrm{LSTM}(r_1, \ldots, r_k; h_0 = u_l) \cdot u_{att}$$

[10] An, Mingxiao, et al. "Neural news recommendation with long-and short-term user representations." Proceedings of the 57th annual meeting of the association for computational linguistics. 2019.
[11] Hochreiter, Sepp, and Jürgen Schmidhuber. "Long short-term memory." Neural computation 9.8 (1997): 1735-1780.

Co-NAML-LSTUR

Click Predictor & Model training



Estimate the **probability** that a **user** will **click on a candidate news** item.

Neural Network-Based Predictor. Use a feedforward neural network (DNN), inspired by the DKN [12].

$$\hat{y} = G([u; r_c])$$

Model Training

Noise Contrastive Estimation (NCE) loss. Calculate based on the ***Pseudo-rank score*** [13] and ***Cross-entropy*** to determine the error between **the news clicked by the user** ($\hat{y}_i^{+}$) and K **news items not clicked by user** ($\bar{\hat{y}}_1 + \bar{\hat{y}}_2 + \cdots + \bar{\hat{y}}_k$)

$$p_i = \frac{exp(\hat{y}_i^{+})}{exp(\hat{y}_i^{+}) + \Sigma_{j=1}^{K} exp(\hat{y}_{i,j}^{-})} ; \mathcal{L} = -\Sigma_{i \in S} \log(pi)$$

[12] Wang, Hongwei, et al. "DKN: Deep knowledge-aware network for news recommendation." Proceedings of the 2018 world wide web conference. 2018.
[13] Qi, Fanchao, et al. "Quoter: A benchmark of quote recommendation for writing." arXiv preprint arXiv:2202.13145 (2022).

MIND-tiny

Motivation and Construction of MIND-Tiny

Statistic	MIND-small	MIND-large
#News	65,238	161,013
#Categories	18	20
#Impressions	230,117	15,777,377
#Clicks	347,727	24,155,470

Table 1: Statistics of the original MIND dataset. *Table adapted from studies* (Wu et al. 2020). [14]

Motivation

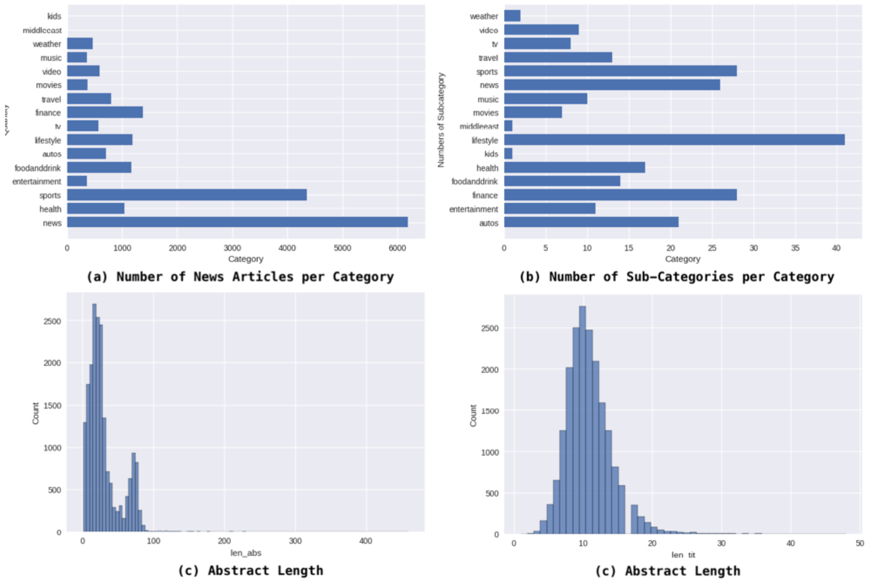
➥ **MIND** dataset is highly comprehensive, its **size poses significant challenges** for **training** on resource constrained device.

➥ **Preserve temporal behavioral** patterns of users.

➥ **Retain representative** and **frequently interacted** news **content**.

Statistic	Value
#News	19,557
#Users	4,844
#Categories	16
#Subcategories	23
Avg. Title Length	10.88 words
Avg. Abstract Length	30.02 words

Table 2: Statistics of the MIND-tiny dataset.



(a) Number of News Articles per Category

(b) Number of Sub-Categories per Category

(c) Abstract Length

(c) Abstract Length

- **High-activity users:** We select users who have exhibited a minimum of **24 click interactions**.
- **Popular news filtering:** We retain those that were clicked by a large number of users (on average more than **150 times**), sorted in descending of popularity.

➥ **Training dataset**: MIND-tiny + MIND-small

➥ **Evaluation dataset**: MIND-small + MIND-large

[14] Wu, Fangzhao, et al. "Mind: A large-scale dataset for news recommendation." Proceedings of the 58th annual meeting of the association for computational linguistics. 2020.

Experiments

Main Results

Table 3: Performance of different methods on MIND-small and MIND-large. The best results are highlighted in bold, second is underlined (higher is better). §: Results from (Li et al. 2022). ‡: MIND-large results from (Wu et al. 2020). †: Results from (Qi et al. 2021).

		MIND-small				MIND-large			
Model	**#Params**	AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
LibFM§	-	0.5974	0.2633	0.2795	0.3429	0.6185	0.2945	0.3145	0.3713
DeepFM§	-	0.5989	0.2621	0.2774	0.3406	0.6187	0.2930	0.3135	0.3705
NRMS‡	22M	0.6183	0.2753	0.2980	0.3653	0.6776	0.3305	0.3594	0.4163
Hi-Fi Ark	21.9M	0.6049	0.2647	0.2927	0.3560	-	-	-	-
NPA§	-	0.6465	0.3001	0.3314	0.3947	0.6592	0.3207	0.3472	0.4037
TANR	21.7M	0.6338	0.2868	0.3169	0.3804	-	-	-	-
LSTUR‡	71.6M	0.6372	0.2769	0.2559	0.3099	0.6773	0.3277	0.3559	0.4134
NAML‡	22.2M	0.6195	0.2528	0.2675	0.3356	0.6686	0.3249	0.3524	0.4091
DKN‡	22.9M	0.5954	0.2659	0.2909	0.3518	0.6460	0.3132	0.2909	0.3518
HieRec†	-	0.6795	0.3287	0.3636	0.4253	0.6903	0.3389	0.3384	0.3948
MINER§	-	**0.6961**	**0.3397**	**0.3762**	**0.4390**	**0.7151**	**0.3618**	**0.3972**	**0.4534**
Co-NAML-LSTUR	46.4M	0.6571	0.3119	0.3465	0.4028	0.6931	0.3420	0.3698	0.4275

- Confirm the Effectiveness of our **efficiency focused** hybrid model, which combines **multi-view news modeling** with **dual-scale user representations**.
- **Comparison with State-of-the-Art Models.** Our models achieves designed for **training** with Fewer resources compared to state-of-the-art methods (SOTA).

Experiments

Ablation Study & Effectiveness of Click Prediction

Model	AUC	MRR	nDCG@10
Co-NAML-LSTUR (Full)	**0.6571**	**0.3119**	**0.4028**
w/o Category Embedding	0.6400	0.2980	0.3870
w/o Word Embedding	0.6105	0.2760	0.3600
Co-NAML-LSTUR w/o BERT	**0.6002**	**0.2529**	**0.3429**
w/o Category Embedding	0.5875	0.2450	0.3305
w/o Word Embedding	0.5650	0.2320	0.3100
NAML + BERT (Baseline)	**0.6350**	**0.2850**	**0.3750**

Table 4: Ablation results on MIND-small. Removing individual components consistently degrades performance across all metrics.

Model	AUC	MRR	nDCG@10
NAML w/o Neural Network	0.6195	0.2528	0.3356
NAML with Neural Network	0.6402	0.2858	0.3819
Co-NAML-LSTUR w/o Neural Network	0.5986	0.2513	0.3406
Co-NAML-LSTUR with Neural Network	**0.6571**	**0.3119**	**0.4028**

Table 5: Effect of click prediction module design on recommendation performance. The neural predictor consistently outperforms the dot-product across all evaluation metrics.

2 × higher training time to compared to the **dot-product variant.**

Confirming that the **complementary role** of **BERT embeddings** when used in a **multi-view encoding pipeline**

Confirming that a learnable **neural click predictor enhances click prediction quality** and **overall recommendation performance** on **our model.**

[15] Pennington, J.; Socher, R.; and Manning, C. D. 2014. Glove: Global vectors for word representation. In Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP), 1532–1543.

Experiments

Case study

News Click Overtimes

Category	Sub-Category	Title	Abstract
movies	movies-celebrity	Jason Momoa Teases 'Way ...	The actor also gushed about...
lifestyle	lifestylebuzz	This Artist Reimagined Disney...	Princess Ariel as Pennywise?
foodanddrink	foodnews	Jennifer Lawrence Hired A Food...	Kinda mad I wasn't invited, TBH.
movies	movies-celebrity	Bruce Willis brought Demi Moore...	Demi wasn't sure how her ex...
tv	tv-celebrity	Joshua Jackson gets backlash...	Joshua Jackson is under fire...
news	elections-2020-us	The Woman Who Flipped Off ...	The cyclist who lost her job...
lifestyle	lifestyledidyouknow	Behind-the-scenes facts about...	The iconic children's show is...
news	newsus	Famed Hollywood Boulevard...	He was the Walk of Fame...
autos	autosenthusiasts	State Trooper Stops Banana Car...	Is it a crime to be this...

Co-NAML-LSTUR			
Category	**Sub-Category**	**Title**	**Abstract**
tv	humor	Chrissy Teigen Scares John Lege...	The model says that scaring ...
news	newsus	Disney's Hulu is raising prices for...	Hulu's live TV bundle for cord...
movies	movienews	13 Reasons Why's Christian Na...	13 Reasons Why's Christian N...
tv	tv-celebrity	Jane Fonda Avoids Fifth Arrest...	This week, the actress was...
tv	tv-celebrity	Joshua Jackson and Jodie Turner...	Joshua Jackson and Jodie ...

The user clicked on one of the recommended articles **(highlighted in green)*

The model accurately prioritizes news from categories previously clicked by the user, such as ***tv-celebrity*** and ***movies.***

Thank You!

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